

Aditya Patra

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EDUCATION

Kalinga Institute of Industrial Technology (KIIT)

Bachelor of Technology in Computer Science and Engineering; CGPA: 8.05/10

Bhubaneswar, Odisha

Expected 2029

PROJECTS

CloudArena | *Kubernetes, Docker, Python, Redis, React, Nginx*

[GitHub](#)

- Built a local Kubernetes-based incident-response environment that simulates infrastructure failures and tests automated remediation workflows.
- Worked with Docker/k3d, Kubernetes workloads, microservices, Redis, Nginx and automated failure scenarios to model cloud infrastructure operations.
- Implemented an AI-powered incident mentor and security-oriented request verification using HMAC-SHA256 within the platform.

TAARAK | *Flutter, Dart, Firebase, Drift/SQLite, Offline-First*

[GitHub](#)

- Developed the application as part of a Smart India Hackathon team, independently implementing the major application functionality using Flutter and Dart.
- Built an offline-first architecture using Drift/SQLite with Firebase authentication and cloud synchronization for reliable operation under limited connectivity.
- Implemented map-based workflows, role-based functionality and automated testing across the application; the project repository documents 440 passing tests.

LLM / RAG Knowledge System | *Python, PyTorch, Transformers, Embeddings, Vector Database*

- Built a practical retrieval-augmented generation pipeline over a curated collection of programming and technical books.
- Processed documents, generated embeddings, stored vector representations in a vector database, and retrieved relevant context for local LLM inference.
- Developed practical understanding of Transformers, tokenization, embeddings, attention, causal language models, model loading, inference, RAG, LoRA/PEFT, QLoRA and quantization.

Text-to-Video Generator | *Python, AI/ML, PDF Processing, TTS, Generative AI*

[GitHub](#)

- Built an AI-assisted pipeline that extracts content from PDFs, summarizes material, generates narration and visual assets, and assembles the output into video.
- Integrated multiple AI processing stages into a single Python workflow for automated content generation.

CERTIFICATIONS & ACTIVITIES

Certified Kubernetes Administrator (CKA)

Feb 2026

KodeKloud

- Hands-on Kubernetes labs covering cluster setup, workload management, networking, storage, security and troubleshooting.

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

Oct 2025

Oracle

Technical Communities & Hackathons

2025–Present

GFG @ KIIT; Google Developer Group (GDG) @ KIIT

- Participated in Smart India Hackathon 2025 internal hackathon, Build with Gemini, HackDay – Hack with Gemini, Dark Route: Trail of Secrets, and other internal technical hackathons.
- Currently exploring open-source contribution and collaborative software development.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Dart, SQL

Cloud & DevOps: Docker, Kubernetes, k3d, Git, GitHub, Nginx, Redis, Cloud Fundamentals

ML / AI: PyTorch, Hugging Face Transformers, Embeddings, RAG, Model Inference, SFT, LoRA/PEFT, QLoRA, Quantization

Databases: PostgreSQL, SQL, SQLite, Drift, Firebase

Frameworks & Tools: Flutter, Firebase, Git, Linux, REST APIs